Linear Algebra 2025 Midterm Solutions

True or False Questions

Answer 'T(rue)' or 'F(alse)'. (You don't have to give a reason, but it might get you half a point if you are wrong.)

1. If rows 1 and 3 of B are the same, then rows 1 and 3 of AB are. F
2. If $(3, 4, 5)$ and $(1, 2, 0)$ are both solutions to $Mx = b$ then M is singular. T
3. If the rows of A are dependent, then so are the columns. F
4. $((AB)^{-1})^T = (B^{-1})^T(A^{-1})^T$ F
5. The matrix $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ has rank 2. T: Two pivots
6. The set of vectors $b = (b_1, b_2, b_3)$ in $\mathbb{R}^3$ with $b_1 - b_2 + 3b_3 = 0$ is a subspace of $\mathbb{R}^3$. T– It is the nullspace of the matrix $[1, -1, 3]$.
7. If A is invertible, then its nullspace is empty. F –The nullspace always contains the zero vector.
8. If $A^T = -2A$ then the rowspace of A equals the column space of A. T
9. If $\{v_1, v_2\}$ is an orthogonal basis of W then so is $\{v_1, cv_2\}$ for $c \neq 0$. T
10. For spaces S and T, if $S = T^\perp$ then $T = S^\perp$. T: $S = (S^\perp)^\perp$.

1. Find a point (x, y, z) with $z = 3$ on the intersection of the planes $x + y + 3z = 6$ and $x - y + z = 4$.

Solution

The proper way to do this might be to solve $\left[\begin{smallmatrix} 1 & 1 & 3 \\ 1 & -1 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} x \\ y \\ z \end{smallmatrix}\right] = \left[\begin{smallmatrix} 6 \\ 4 \end{smallmatrix}\right]$ and then set the variable z (by which the solution would be parameterised, as the third column will be free) to be 3.
But if you set $z = 3$ to start you get a smaller system to solve: $\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right] = \left[\begin{smallmatrix} -3 \\ 1 \end{smallmatrix}\right]$.
Solving this:

$$\left[\begin{array}{cc|c} 1 & 1 & -3 \\ 1 & -1 & 1 \end{array}\right] \longrightarrow \cdots \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & -2 \end{array}\right]$$

yields the solution $(x, y, z) = \boxed{\text{(-1,-2,3)}}$.

2. Find a, b and c where

$$\begin{bmatrix} a & 2 \\ 1 & 1 \\ b & 4 \end{bmatrix} \begin{bmatrix} c \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix}.$$

Solution

From the second row we get that $c = 2$. So $a = -1$ and $b = -3/2$. Thus $\boxed{\text{(a,b,c) = (-1,-3/2,2)}}$.

3. (a) Write out the 3×3 row operation matrix E_1 that adds three copies of row 1 to row 2.
   (b) Write out the 3×3 row operation matrix E_2 that switches row 1 with row 3.
   (c) Compute E_1^n, and E_2^n.

Solution

$E_1 = \left[\begin{smallmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$ and $E_2 = \left[\begin{smallmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{smallmatrix}\right]$. E_1^n adds three copies of row 1 to row 2 n times, so add $3n$ copies of row 1 to row 3: $E_1^n = \left[\begin{smallmatrix} 1 & 0 & 0 \\ 3n & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right]$. E_2^n switches these two rows n times, so is I if n is even E_2 if n is odd.

4. Where the matrix A has the LU decomposition

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix},$$

find another LU decomposition of A.

Solution

There are all sorts of these. The easiest way to do it is as we did in class, extracting the diagonal:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 6 \end{bmatrix}$$

So $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 6 \end{bmatrix}$ is another LU decomposition of A.

5. Find the inverse of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$.

Solution

Probably by Gauss-jordan, you find $\begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

6. The matrix equation $M\mathbf{x} = \mathbf{b}$ has solutions $(1, -1, 2)$ and $(-2, 2, -4)$. What is $\mathbf{b}$?

Solution

It is $\mathbf{0}$. As $2(1, -1, 2) + (-2, 2, -4) = (0, 0, 0)$ is also a solution to $M\mathbf{x} = \mathbf{b}$, we must have $\mathbf{b} = \mathbf{0}$.

7. For the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 & 3 & 0 \\ 0 & 1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

(a) What is the rank of A?

(b) Find the solution of $Ax = (2, 2, 3, 0)^T$

(c) What are the dimensions of the nullspace and the rowspace of A.

Solution

(a) There are three pivot columns, (or clearly three independent rows) so the rank is $\boxed{3}$. (b) Where $x = (x_1, x_2, x_3, x_4, x_5)$ the solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -3 \\ -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \\ 0 \\ 0 \\ 3 \end{bmatrix}.$$

(c) This solution has two indeterminates, so the nullspace has dimension $\boxed{2}$. (Alternately you know the rank, so can compute that it is $n-r = 5-3 = \boxed{2}$.) The rowspace has dimension $\boxed{3}$, same as the rank.

8. Let A and B be $n \times n$ matrices. Show that AB is invertible if and only if A and B are.

Solution

If A and B are invertible, then $B^{-1}A^{-1}$ is the inverse of AB, so AB is invertible. If AB is invertible, then $A(BM) = (AB)M = I_{n\times n}$ for some matrix $M = (AB)^{-1}$. So the n independent columns of $I_{n\times n}$ are in the columnspace of A. Thus A has rank at least, and so equal to, n. An $n \times n$ matrix of rank n is invertible. Similarily $MAB = I_{n\times n}$ shows that B is invertible.

9. Give the matrix M_θ such that $M_\theta v$ is the vector $v \in \mathbb{R}^2$ rotated by θ degrees (radians). Using this matrix, derive the angle sum formula for $\sin(\theta + \gamma)$.

Solution

We have $M_\theta = \left[\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right]$. So $M_{\theta+\gamma}$ is (writing c_θ for $\cos\theta$, etc,)

$$\begin{bmatrix} c_{\theta+\gamma} & -s_{\theta+\gamma} \\ s_{\theta+\gamma} & c_{\theta+\gamma} \end{bmatrix} = \begin{bmatrix} c_\theta & -s_\theta \\ s_\theta & c_\theta \end{bmatrix} \begin{bmatrix} c_\gamma & -s_\gamma \\ s_\gamma & c_\gamma \end{bmatrix} = \begin{bmatrix} c_\theta c_\gamma - s_\theta s_g & * \\ s_\theta c_\gamma + c_\theta s_g & * \end{bmatrix}$$

So $\sin\theta + \gamma = \sin\theta\cos\gamma + \cos\theta + \sin\gamma$.

10. Find a basis for the orthogonal complement $A^\perp$ of the space A spanned by $(1, 2, 2, 3)$ and $(1, 3, 3, 2)$.

Solution

Proper way: $A^\perp$ is the nullspace of $M = \begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix}$ which we find by solving $Mx = 0$ by elimination:

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 1 & 3 & 3 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 1 & 1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 1 & 1 & -1 \end{bmatrix}.$$

The nullspace is $\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 0 \\ -1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix}$, which has a basis $\boxed{\{\ (0,-1,1,0),\ (-5,1,0,1)\ \}}$.

Dirty way: you know that the complement of A has dimension 2, so just find two indepentent vectors that are orthogoal to the two given vectors.